$F_N - T - Mg = 0$ (1)



**3.** A uniform solid disk of radius R and mass M is free to rotate on a frictionless pivot through a point on its rim as shown in the figure. If the disk is released from rest in the position shown by the gray circle, (a) what is the speed of its center of mass when the disk reaches the position indicated by the dashed circle? (b) What is the speed of the lowest point on the disk in the dashed position?
(Moment of inertia of a uniform circular disc about an axis passing through its center is $1/2\ MR^2$)

Yarıçapı R ve kütlesi M olan düzgün katı bir disk, şekilde gösterildiği gibi çevresi üzerindeki bir nokta etrafında dönebilecek şekilde tutturulmuştur. Eğer disk gri daire ile gösterilen konumdan serbest bırakılırsa, (a) kesikli çizgi ile gösterilen konuma geldiğinde kütle merkezinin hızı ne olur? (b) kesikli konumda, disk üzerindeki en alt noktanın hızı nedir?
(Düzgün kütle dağılımına sahip dairesel bir diskin merkezinden geçen eksene göre eylemsizlik momenti $1/2\ MR^2$ ifadesi ile verilir.)

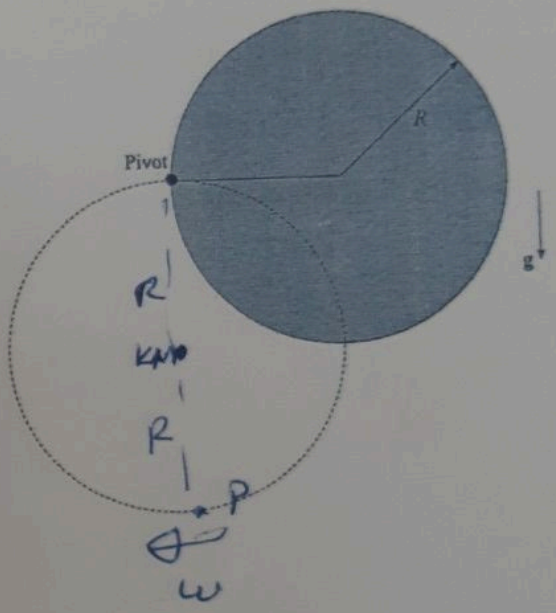
bir konarı etrafında dönen disk için

$$I = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$$

enerjinin korunumun

$$MgR = \frac{1}{2}I\omega^2$$

$$\omega^2 = \frac{2MgR}{I} = \frac{2MgR}{\frac{3}{2}MR^2} = \frac{4g}{3R}$$




$$\omega = \sqrt{\frac{4g}{3R}} = 2\sqrt{\frac{g}{3R}}$$

a) $v_{KM} = r\omega = R \cdot \sqrt{\frac{4g}{3R}} = \sqrt{\frac{4gR}{3}} = 2\sqrt{\frac{gR}{3}}$

($r = R$)

b) $v_P = r\omega = 2R\sqrt{\frac{4g}{3R}} = 2 \cdot 2\sqrt{\frac{gR}{3}} = 4\sqrt{\frac{gR}{3}}$

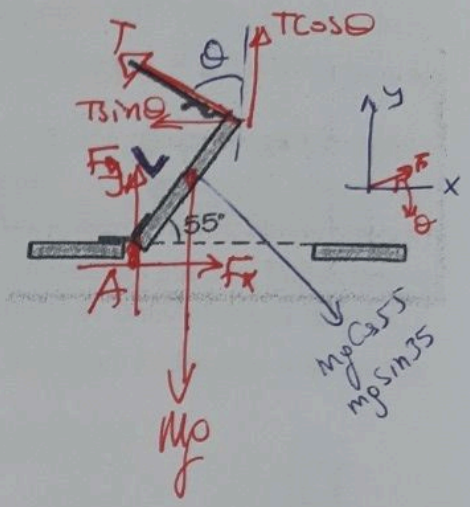
($r = 2R$)

4)- A trapdoor of length and width 2 m and mass 20 kg is hinged at one edge and is held opened at an angle of 55° with respect to the floor by a rope attached at the raised edge of the door as shown in the figure. The rope is perpendicular to the door.

a) What is the tension in the rope ? b) What is the force acting on the door at the hinge?

Kütlesi 20 kg, kenarları 2 m olan kare şeklindeki bir tuzak kapağı bir kenarından menteşelenmiş olup, diğer kenarından bir iple asılmıştır. Tuzağın kapısı yatayla 55° lik açı yapmaktadır. İp kapıya dik konumdadır.

**a)** İpteki gerilme nedir? **b)** Kapıya etki eden kuvvet nedir?(menteşenin olduğu noktada)




(10) a) A noktasına göre toplam tork

$$\sum \tau = Mg\frac{L}{2}\cos\theta - TL = 0$$

$$T = \frac{1}{2} Mp \cos\theta$$

$$T = \frac{1}{2}(20kg)(9,8m/s^2)\underbrace{\cos 55^\circ}_{0,573}$$

$$\boxed{T = 56\,N}$$

$\cos(55) = 0,573$

$\sin(55) = 0,819$

(10) b) x- yönündeki toplam kuvvet sıfırdır.

(2) $F_x - T\sin\theta = 0 \quad F_x = T\sin\theta = (56)\sin 55$

(2) $\boxed{F_x = 46\,N}$

y-yönündeki toplam kuvvet,

(2) $F_y + T\cos\theta - Mp = 0 \Rightarrow F_y = Mp - T\cos\theta$

$F_y = (20kg)(9,8m/s^2) - (56N)\cos(55)$

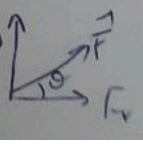
(2) $\boxed{F_y = 164\,N}$

(2) $\boxed{\vec{F} = (46\,N)\hat{i} + (164\,N)\hat{j}}$

veya $|\vec{F}| = \sqrt{(164)^2 + (46)^2}$

$F = 170,47\,N$

$(\tan\theta) = \frac{164}{46} \Rightarrow \boxed{\theta = 74,83^\circ}$

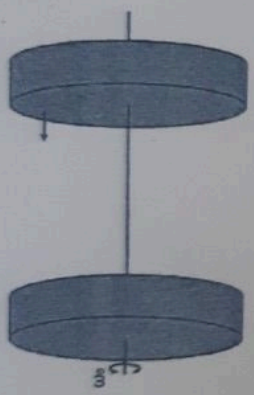


disc falls on top of the first. Because of friction between the disks they come to rotate at the same rate in 0.01 s. (For a disc $I = \frac{1}{2}MR^2$)

(a) What is the moment of inertia of each disc?
(b) What is the final angular speed of the two discs?
(c) What is the initial and final values of the total kinetic energy? Where does the lost kinetic energy go?
(d) What is the average torque acting on the dropped disc? What is the source of this torque?
(e) What is the average total torque acting on the system of the two discs?

Kütlesi 250 kg ve yarıçapı 2,00 m olan bir disk sürtünmesiz bir yüzey üzerinde 4,00 rad/s süratiyle dönmekte iken tam üstüne başlangıçta dönmeyen özdeş bir silindir düşmüştür. İki disk arasındaki sürtünmeden dolayı 0,01 s içinde her ikisi de ortak bir açısal sürate ulaşırlar. (Disk için eylemsizlik momenti $I = \frac{1}{2}MR^2$)

(a) Her bir diskin eylemsizlik momenti nedir?
(b) Disklerin son açısal süratleri nedir?
(c) Toplam kinetik enerjinin ilk ve son değerleri nedir? Kaybolan kinetik enerji nereye gider?
(d) İlkinin üzerine düşen diske etki eden ortalama tork nedir? Bu torkun kaynağı nedir?
(e) İki diskten oluşan sisteme etki eden toplam ortalama tork nedir?



a) $I = \frac{1}{2}MR^2 = \frac{1}{2}(250)(2)^2 = 500\ kg\,m^2$ | 2 points

b) $\tau_{net} = 0$ for the system, L is conserved

$I\omega_0 = 2I\omega \Rightarrow \omega = \frac{\omega_0}{2} = 2.0\ \frac{rad}{s}$ | 4 points

c) $K_0 = \frac{1}{2}I\omega_0^2 = \frac{1}{2}\cdot 500\cdot(4)^2 = 4000\ J$ | 2 points

$K = \frac{1}{2}2I\omega^2 = \frac{1}{2}\cdot 1000\cdot(2)^2 = 2000\ J$ | 2 points

It goes into the internal energy of the discs due to the friction between them. Their temperature should increase. | 2 points

d) $\tau = \frac{\Delta L}{\Delta t} = \frac{I\omega - 0}{\Delta t} = \frac{500\cdot 2}{0.01} = 10^5\ N\cdot m = 100\,000\ N\cdot m$ | 2 points

. It is produced by the friction between the discs. | 2 points

e) Since there are no external forces $\tau_{total} = 0$ for the whole system. | 4 points

3. A particle has a displacement $x = 0.4\cos(3t + \pi/4)$, where $x$ is in meters and $t$ is in seconds.
a) Find the frequency $f$ and period $T$ of the motion.
b) Where is the particle at t = 0 ?
c) Find an expression for the velocity of the particle.
d) What is the velocity at time t = 0?
e) What is the maximum velocity?

Bir parçacığın yer değiştirmesi $x = 0{,}4\cos(3t + \pi/4)$, denklemiyle tanımlanmıştır. Burada $x$ metre ve $t$ saniye olarak tanımlanmıştır.
a) Hareketin frekansı ($f$) ve periyodunu ($T$) bulunuz.
b) t= 0 anında parçacığın konumunu bulunuz.
c) Parçacığın hızını ifade eden denklemi yazınız.
d) t= 0 anında parçacığın hızını bulunuz.
e) Maksimum hızı hesaplayınız.

$X(t) = A\cos(\omega t + \phi)$ , $X(t) = 0{,}4\cos(3t + \frac{\pi}{4})$

$A = 0{,}4\,m \quad \omega = 3\,rad/s \quad , \phi = \frac{\pi}{4}$ 2

a) $f = \frac{\omega}{2\pi} = \frac{3}{2\pi} \Rightarrow \boxed{f = 0{,}477\,Hz}$ 2 $T = \frac{1}{f} = \frac{1}{0{,}477} \Rightarrow \boxed{T = 2{,}09\,s}$

b) $t=0 \quad X(t=0) = (0{,}4m)\cos\left[(3\,rad/s)(0) + \frac{\pi}{4}\right] = (0{,}4m)\cos\frac{\pi}{4}$

$\boxed{X(t=0) = 0{,}283\,m}$ 4

c) $v = \frac{dx}{dt} = \frac{d}{dt}\left[0{,}4\cos(3t + \frac{\pi}{4})\right] = -(1{,}2m)\cdot\sin(3t + \frac{\pi}{4})$

$\boxed{v(t) = -(1{,}2m)\sin(3t + \frac{\pi}{4})}$ 4

d) $v(t=0) = -(1{,}2m)\sin(0 + \frac{\pi}{4}) \Rightarrow \boxed{v(t=0) = -0{,}849\,m/s}$ 4

e) $\boxed{v_{max} = 1{,}2\,m/s}$ 4

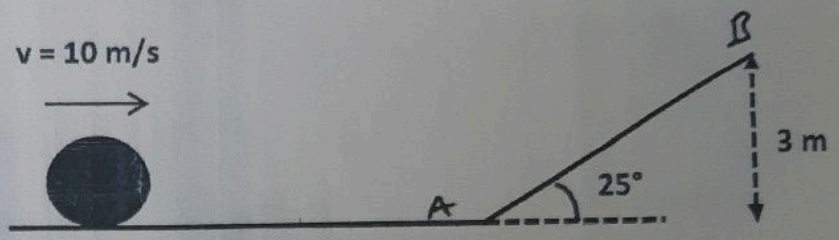
translational speed of 10 m/s, radius of 0.2 m and mass of 25 kg. The sphere approaches a 25° inclined plane of height 3 m and rolls up the plane without slipping. ($I_{CM} = 2/5Mr^2$)

a) Calculate the **kinetic energy** of the sphere as it rolls along the horizontal surface.
b) Calculate the **magnitude of the sphere's velocity** just as it leaves the top of the incline.

Katı küresel bir top şekildeki gibi yatay yüzeyde kaymadan yuvarlanmaktadır. Kürenin 10 m/s lik sabit bir ötelenme suratı olup yarıçapı 0,20 m, kütlesi ise 25 kg dır. Küre yüksekliği 3 m olan bir 25 lik bir eğik düzleme yaklaşmakta ve eğik düzlemden yukarı doğru yine kaymadan yuvarlanarak çıkmaktadır ($I_{KM} = 2/5Mr^2$).

a)- Top yatay düzlemde hareket ederken **kinetik enerjisini** bulunuz.

b)- Top eğik düzlemi tam terk ederken **hızının büyüklüğünü** bulunuz.




a)

$$E_A = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \qquad u = \omega r \;,\; \omega = \frac{u}{r}$$

$$= \frac{1}{2}(25\,kg)(10\,m/s)^2 + \frac{1}{2}\left[\frac{2}{5}\cdot 25\,(0.2)^2\right]\cdot\frac{(10\,m/s)^2}{(0.2)^2}$$

$$\boxed{E_A = 1750\ \text{Joule}}$$

b) $E_B = K_B + U_B$ , $U_B = mgh = (25\,kg)(9.8\,m/s^2)(3m)$

$E_B = E_A$ $\qquad U_B = 735$ Joule

$$K_B = E_B - U_B, = E_A - U_B = 1750 - 735 = 1015\ \text{Joule}$$

$$1015\ \text{Joule} = \frac{1}{2}mv_B^2 + \frac{1}{2}\left[\frac{2}{5}mr^2\right]\left(\frac{v_B}{r}\right)^2 = \frac{2}{5}\cdot 1015$$

$$= \frac{1}{2}25\,v_B^2 + \frac{1}{2}\frac{2}{5}25\,v_B^2 = \underbrace{\left(\frac{25}{2}+5\right)}_{17.5}v_B^2$$

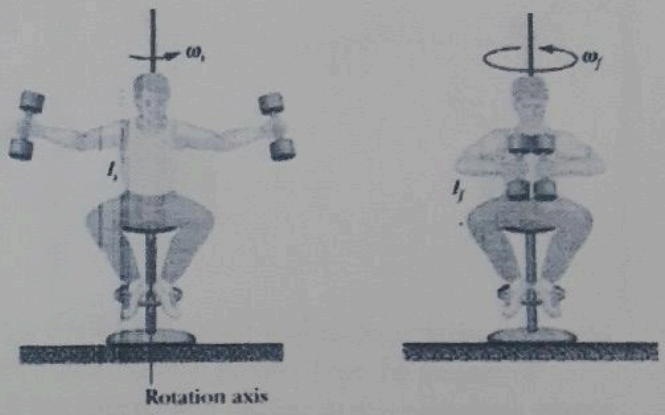
$$v_B^2 = \frac{1015}{17.5} = 58 \Rightarrow \boxed{v_B = 7.62\ m/s}$$

4)- A student stands at the center of a frictionless turntable, holding his arms extended horizontally with a 5.0 kg dumbbell in each hand. He is set rotating about the vertical axis, making one revolution in 1.0 s. His moment of inertia (with the dumbbells) is 10.0 kg m$^2$ ($I_i$) with arms outstretched and 5.0 kg m$^2$ $\left(I_f = \frac{1}{2} I_i\right)$ with his hands at his stomach.

a) Find his final angular velocity if he pulls the dumbbells in to his stomach.

b) Find the difference of kinetic energy

Bir öğrenci sürtünmesiz olarak dönen bir taburenin ortasında durmaktadır. Kollarını açmış ve her bir elinde 5,0 kg kütleli dambıl vardır. Dikey eksen etrafında dönerken, bir deviri 1,0 s de tamamlamaktadır. Öğrencinin başlangıçtaki eylemsizlik momenti (Kolları açık ve ellerinde dumbıllar varken) 10,0 kg m$^2$ ($I_i$) dir. Kollarını karnına çektikten sonraki eylemsizlik momenti 5,0 kg m$^2$ dir $\left(I_f = \frac{1}{2} I_i\right)$.




a) Öğrencinin dumbılları karnına doğru çektikten sonraki açısal hızını bulunuz.

b) Kinetik enerji farkını bulunuz.

a) 10

$L_i = L_f$.

$I_i \omega_i = I_f \omega_f \Rightarrow \omega_f = \left(\frac{I_i}{I_f}\right) \omega_i = 2\omega_i$ (4)

$\omega_i = 1 \text{ dev/s} = 1 \cdot 2\pi \text{ rad/s}$ $\boxed{\omega_i = 2\pi \text{ rad/s}} = 6{,}28 \text{ rad/s}$ (3)

$\omega_f = 2\omega_i = 2 \cdot 2\pi \Rightarrow \boxed{\omega_f = 4\pi \text{ rad/s}} = 12{,}56 \text{ rad/s}$ (3)

b) 10

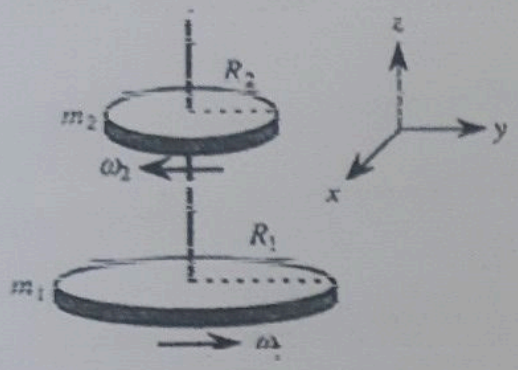
$K_i = \frac{1}{2} I_i \omega_i^2 = \frac{1}{2} 10 (2\pi)^2 = 20\pi^2 \text{ J}$ (4)

$K_f = \frac{1}{2} I_f (\omega_f)^2 = \frac{1}{2} 5 (4\pi)^2 = 40\pi^2 \text{ J}$ (4)

$\Delta K = K_f - K_i = 20\pi^2 \text{ Joule} = 197{,}19 \text{ Joule}$ (2)

**PROBLEM (7) (15 points)**

Two solid disks with masses $m_1$ and $m_2$ are rotating freely without friction with angular velocities $\omega_1$ and $\omega_2$, respectively, about the same vertical axle passing through their centers of mass as shown. As indicated, $\omega_1$ and $\omega_2$ are in opposite directions. The upper disk is dropped down the axle to land on and couple (because of friction) to the lower disk. The disks rub against each other and finally reach a common final angular velocity $\omega_f$. Express your results in terms of parameters given in the figure. (For a solid disk about the central axis: $I = MR^2/2$)




(a) Calculate the initial angular momentum vector $\vec{L}$ of the two-disk system before the disks collide.

$$\vec{L} = \vec{L}_1 + \vec{L}_2$$

Use the right hand rule and take upward direction positive.

$$\therefore \vec{L}_1 = +I_1\omega_1\hat{k} = \frac{1}{2}m_1R_1^2\omega_1\hat{k}$$

$$\vec{L}_2 = -I_2\omega_2\hat{k} = -\frac{1}{2}m_2R_2^2\omega_2\hat{k}$$

$$\therefore \boxed{\vec{L} = \frac{1}{2}(m_1R_1^2\omega_1 - m_2R_2^2\omega_2)\hat{k}}$$

(b) Find the common final angular speed $\omega_f$ of the system.

For the two-disk system: $\sum \vec{\tau}_{ext} = 0 \longrightarrow \vec{L}_i = \vec{L}_f$

$$\frac{1}{2}(m_1R_1^2\omega_1 - m_2R_2^2\omega_2)\hat{k} = \frac{1}{2}(m_1R_1^2 + m_2R_2^2)\omega_f\hat{k}$$

$$\therefore \boxed{\omega_f = \frac{m_1R_1^2\omega_1 - m_2R_2^2\omega_2}{m_1R_1^2 + m_2R_2^2}}$$

(c) How much kinetic energy is lost in this situation?

$$K_i = \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 = \frac{1}{2}\left(\frac{1}{2}m_1R_1^2\omega_1^2 + \frac{1}{2}m_2R_2^2\omega_2^2\right)$$

$$K_f = \frac{1}{2}I_{tot}\omega_f^2 = \frac{1}{2}\left(\frac{1}{2}m_1R_1^2 + \frac{1}{2}m_2R_2^2\right)\omega_f^2$$

$$\Delta K = K_f - K_i = \frac{1}{4}(m_1R_1^2 + m_2R_2^2)\left(\frac{m_1R_1^2\omega_1 - m_2R_2^2\omega_2}{m_1R_1^2 + m_2R_2^2}\right) - \frac{1}{4}(m_1R_1^2\omega_1^2 + m_2R_2^2$$

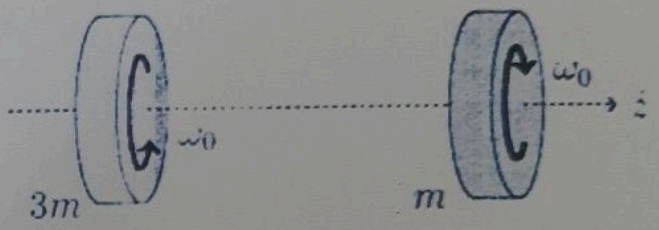
3)-

Two homogenous disks have radius $R$ but masses $3m$ and $m$. They are aligned on the $\hat{z}$ axis as shown in the figure and are initially rotating around their respective center of masses **in opposite directions** with the same angular speed $\omega_0$. The two disks are moving slowly towards each so that when they touch friction force will slow them down until both of them are rotating together as a single body. Ignore any gravitational effects. $\left(I_{disk} = \frac{1}{2}MR^2\right)$

a) Calculate the initial angular momentum vector $\vec{L}$ of the two-disk system bofore the disks collide.
b) Find the final angular velocity $\omega_f$ of the two-disk system after they are stuck together.

İki homojen diskin yarıçapları $R$, kütleleri $3m$ ve $m$ olarak verilmiştir. Şekilde gösterildiği gibi $\hat{z}$ ekseni etrafında, **ters yönlerde** $\omega_0$ açısal hızıyla dönmektedirler. İki disk yavaşça birbirlerine doğru hareket ediyor ve temas ettikten sonra kendi aralarındaki sürtünmeden dolayı yavaşlayıp bir süre sonra tek bir cisim gibi dönmeye devam ediyorlar. Gravitasyonel etkileri ihmal ediniz. $\left(I_{disk} = \frac{1}{2}MR^2\right)$

a) İki diskten oluşan sistemin çarpışmadan önceki açısal momentum vektörünü $\vec{L}$ hesaplayınız.
b) İki diskin bir araya gelerek tek bir cisim gibi hareket etmesiyle oluşan sistemin $\omega_f$ açısal hızını bulunuz.




a) $\vec{L} = I_1\vec{\omega}_1 + I_2\vec{\omega}_2 \;,\; \Rightarrow \vec{L} = (I_1 - I_2)\omega_0\hat{z} \;,\; I_1 = \frac{1}{2}(3m)R^2$
$I_2 = \frac{1}{2}mR^2$

b) $\vec{L}_i = \vec{L}_f$

$$I_1\omega_0 - I_2\omega_0 = (I_1 + I_2)\omega_f$$

$$\omega_f = \frac{I_1\omega_0 - I_2\omega_0}{I_1 + I_2} = \frac{I_1 - I_2}{I_1 + I_2}\omega_0$$

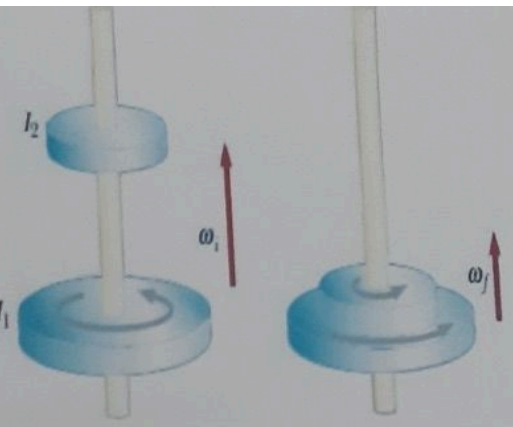
$$\omega_f = \frac{\frac{3}{2}mR^2 - \frac{1}{2}mR^2}{\frac{3}{2}mR^2 + \frac{1}{2}mR^2}\omega_0 = \frac{3-1}{3+1}\omega_0$$

$$\boxed{\omega_f = \frac{\omega_0}{2}}$$

7) Eylemsizlik momenti ihmal edilebilir bir şaft üzerinde bir tekerlek şekil'deki gibi 900 devir/dak açısal hız ile dönüyor. Başlangıçta hareketsiz olan ve eylemsizlik momenti birincisinin iki katı olan ikinci bir tekerlek aynı şafta bağlanıyor.

**a)** İki tekerlek ve şafttan oluşan sistemin açısal hızı nedir?

**b)** Sistemde oluşan dönme kinetik enerjisindeki değişimi bulunuz.




$\omega_1 = 900$ devir/dak , $I_2 = 2I_1$

**a)** Sistem üzerine etkiyen net tork sıfır olduğundan sistemin açısal momentumu korunur ($\vec{L}$ = sabit).

Başlangıçta açısal momentum, $L_1 = I_1 \omega_1$

Sonraki " " , $L_2 = (I_1 + I_2)\omega_2$

$$\omega_2 = \frac{I_1}{I_1 + I_2}\omega_1 = \frac{I_1}{I_1 + 2I_1}\omega_1 = \frac{\omega_1}{3} = \frac{900}{3} = 300 \frac{dev}{dak}$$

$$\omega_2 = 300 \frac{dev}{dak} = \frac{300 \times 2\pi}{60} = 10\pi \frac{rad}{s}$$

**b)** Başlangıçta kinetik enerji, $K_1 = \frac{1}{2} I_1 \omega_1^2$

Sonraki " " , $K_2 = \frac{1}{2}(I_1 + I_2)\omega_2^2$

Kinetik enerjideki değişim,

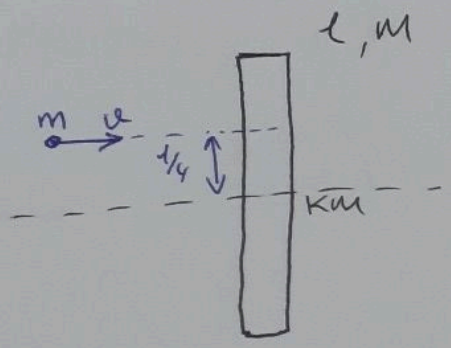
$$\Delta K = K_2 - K_1 = \frac{3}{2} I_1 \omega_2^2 - \frac{1}{2} I_1 \omega_1^2 = \frac{3}{2} I_1 \left(\frac{\omega_1}{3}\right)^2 - \frac{1}{2} I_1 \omega_1^2$$

$\Delta K = -\frac{2}{3} K_1$ Sistem başlangıçtaki enerjisinin $\frac{2}{3}$'ni kaybeder.

# Örnek

Çubuk yatay sürtünmesiz düzlemde duruyor.
m cismi v hızıyla gidip çarpıyor ve v/2 hızıyla geri dönüyor.

$I_{km} = \frac{1}{12} ML^2$




a) Çubuğun $V_{km} = ?$

b) $\omega_{sonrası} = ?$

c) ~~Kaybolan enerjinin~~ Çarpışmada bağlanpıçıklı enerjinin % kaçı kaybolmuştur.

a) $p_{ö} = mv$ , $p_{sonra} = -m\frac{v}{2} + M V_{km}$

$$mv = -m\frac{v}{2} + M V_{km} \Rightarrow \frac{3}{2}mv = MV_{km} \quad V_{km} = \frac{3mv}{2M}$$

b) $L_{ö} = mv\frac{l}{4}$ , $I_{km} = \frac{1}{12} ML^2$

$$L_{son} = -m\left(\frac{v}{2}\right)\frac{l}{4} + I\omega_{km}$$

$$mv\frac{l}{4} = -m\frac{v}{2}\frac{l}{4} + I\omega \Rightarrow mvl\left(\underbrace{\frac{1}{4} + \frac{1}{8}}_{3/8}\right) = I\omega$$

$$\omega = \frac{mvl\,3/8}{I} = \frac{3/8\, mvl}{\frac{1}{12}ML^2} = \frac{3}{8}\cdot\frac{12}{3}\frac{mvl}{ML^2} \Rightarrow \omega = \frac{9mv}{2ML}$$

$$\boxed{\omega = \frac{9mv}{2ML}}$$

c) $K_i = \frac{1}{2}mv^2$ , $K_{son} = \frac{1}{2}m\left(\frac{v}{2}\right)^2 + \frac{1}{2}MV_{km}^2 + \frac{1}{2}I\omega^2$

$$= \frac{1}{2}m\left(\frac{v}{2}\right)^2 + \frac{1}{2}M\left(\frac{3mv}{2M}\right)^2 + \frac{1}{2}\left(\frac{1}{12}ML^2\right)\left(\frac{9mv}{2ML}\right)^2$$

$$K_{son} = \left(\frac{1}{4} + \frac{63m}{16M}\right)\frac{1}{2}mv^2 = \frac{1}{8}\left(1 + \frac{63m}{4M}\right)mv^2$$

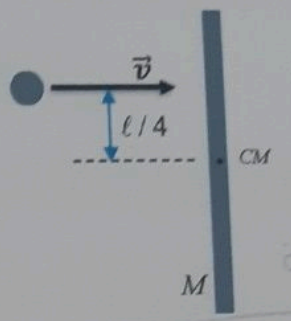
$$\frac{\Delta K}{K_i} = \frac{K_{son} - K_i}{K_i} \Rightarrow K_{son} - K_i = \Delta K = \frac{1}{8}\left(1 + \frac{63m}{4M}\right)mv^2 - \frac{1}{2}mv^2$$

$$= -\frac{3}{8}\left(1 - \frac{21m}{4M}\right)mv^2.$$

=

**4.** A thin rod of mass $M$ and length $\ell$ rests on a frictionles icy surface and is struck at a point $\ell/4$ from its CM by a hockey puck of mass $m$ moving at speed $v$. It is observed that after the collision the puck recoils with a speed $v/2$ while the rod starts to move to the right rotating about its center of mass at the same time.

($I_{CM} = \frac{1}{12} M\ell^2$ for the rod)




(a) (7 Pts.) Find the velocity of the center of mass of the rod after the collision.

(b) (7 Pts.) Find the angular velocity of the rod about its center of mass after the collision.

(c) (6 Pts.) Find the change in the energy of the system during the collision.

**Solution**: Linear momentum and angular momentum about the center of mass of the of the rod is conserved during the collision. (a) Conservation of linear momentum implies

$$mv = -m\frac{v}{2} + MV_{CM} \rightarrow V_{CM} = \frac{3mv}{2M}.$$

$$\boxed{V_{CM} = \frac{3mv}{2M}}$$

(b) Conservation of angular momentum about the CM of the rod implies

$$mv\frac{\ell}{4} = -m\frac{v}{2}\frac{\ell}{4} + \left(\frac{1}{12}M\ell^2\right)\omega \rightarrow \omega = \frac{9mv}{2M\ell}.$$

$$\boxed{\omega = \frac{9mv}{2M\ell}}$$

(c) Kinetic energy of the system before the collision is $K_i = \frac{1}{2}mv^2$. After the collision both the recoiling puck and the rod has kinetic energy. Kinetic energy of the rod has both translartional and rotational parts. Hence $K_f = \frac{1}{2}m\left(\frac{v}{2}\right)^2 + \frac{1}{2}MV_{CM}^2 + \frac{1}{2}I_{CM}\omega^2$. Using the results found, we have

$$K_f = \frac{1}{2}m\left(\frac{v}{2}\right)^2 + \frac{1}{2}M\left(\frac{3mv}{2M}\right)^2 + \frac{1}{2}\left(\frac{1}{12}M\ell^2\right)\left(\frac{9mv}{2M\ell}\right)^2 \rightarrow K_f = \frac{1}{8}\left(1+\frac{63m}{4M}\right)mv^2.$$

Hence, $\Delta K = \frac{1}{8}\left(1+\frac{63m}{4M}\right)mv^2 - \frac{1}{2}mv^2 = -\frac{3}{8}\left(1-\frac{21m}{4M}\right)mv^2$.

$$\boxed{\Delta K = -\frac{3}{8}\left(1-\frac{21m}{4M}\right)mv^2}$$

85. A spool of wire of mass $M$ and radius $R$ is unwound under a constant force $\mathbf{F}$ (Fig. P10.85). Assuming the spool is a uniform solid cylinder that doesn't slip, show that (a) the acceleration of the center of mass is $4\mathbf{F}/3M$ and (b) the force of friction is to the *right* and equal in magnitude to $F/3$. (c) If the cylinder starts from rest and rolls without slipping, what is the speed of its center of mass after it has rolled through a distance $d$?

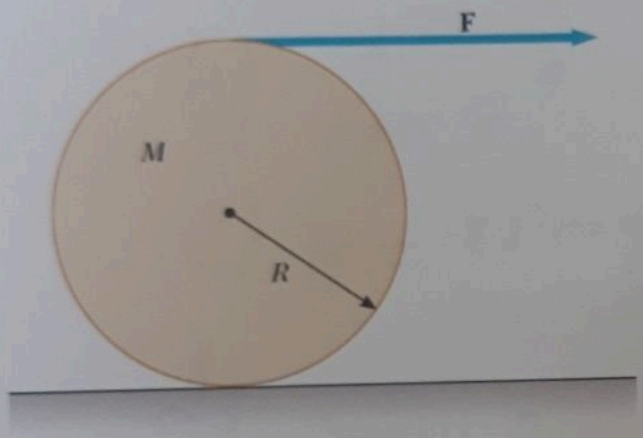



**Figure P10.85**

P10.85 (a) $\sum F_x = F + f = Ma_{CM}$

$\sum \tau = FR - fR = I\alpha$

$FR - (Ma_{CM} - F)R = \dfrac{Ia_{CM}}{R}$ $\qquad \boxed{a_{CM} = \dfrac{4F}{3M}}$

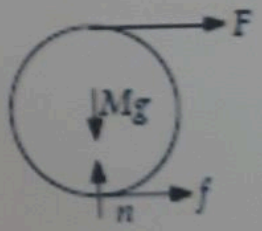



FIG. P10.85

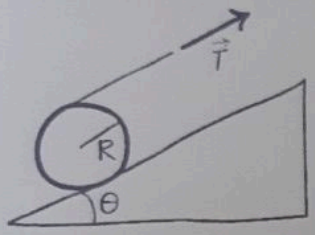
(b) $f = Ma_{CM} - F = M\left(\dfrac{4F}{3M}\right) - F = \boxed{\dfrac{1}{3}F}$

(c) $v_f^2 = v_i^2 + 2a(x_f - x_i)$

$v_f = \boxed{\sqrt{\dfrac{8Fd}{3M}}}$

**5.** Kütlesi $M$, yarıçapı $R$, ve kendi eksenine göre eylemsizlik momenti $I$ olan bir silindir, eğim açısı θ olan bir eğik düzlemin üzerinde durmaktadır. Bir sicim, silindirin etrafına dolanmış ve eğik düzleme paralel $T$ gerilmesi ile çekilmektedir. Statik sürtünme katsayısı kaymadan yuvarlanmayı sağlayacak şekilde yeterince büyüktür. (a) Silindiri dengede tutmak için gerekli $T_0$ gerilmesini bulunuz. (b) $T_0$'dan farklı belli bir $T$ gerilmesi için silindirin ivmesi ne olur.

**5.** A cylinder of known mass $M$, radius $R$, and rotational inertia $I$ (with respect to its own axis) is placed on an inclined plane with an angle $\theta$. A string is wound around the cylinder and pulled with a tension $T$ parallel to the inclined plane. The coefficient of static friction is large enough to prevent slipping. (a) Find the tension $T_0$ needed to keep the cylinder in equilibrium. (b) Find the acceleration of the cylinder if the tension is known and is different from $T_0$.

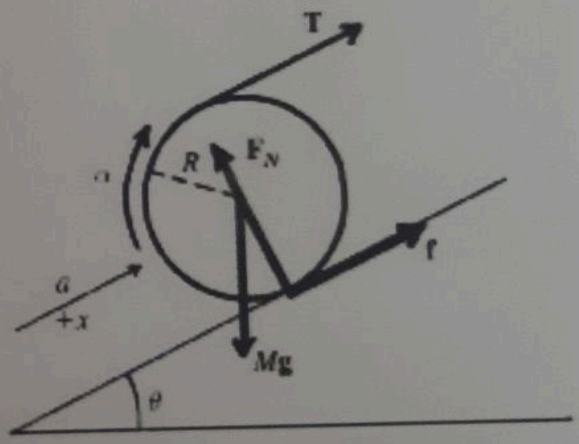





***Çözüm***

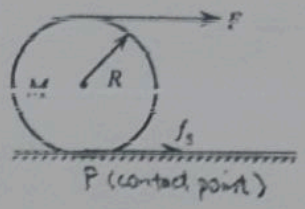
(a) Silindir denge konumunda $\sum \tau = 0$.
Tork kütle merkezine göre hesaplanır
Şekildeki kuvvet diyagramından
$T_0 R - fR = 0 \Rightarrow T_0 = f$
$\sum F_x = 0$ sağlanır. Buna göre
$f + T_0 - Mg\sin\theta = 0$, buradan
denge konumundaki $T_0$ gerilimi $\boxed{T_0 = \frac{1}{2}Mg\sin\theta}$

(b) Farklı bir $T$ gerilimi için hareket denklemi
$f + T - Mg\sin\theta = Ma$
Dönme hareket denklemi
$\sum \tau = I\alpha$, $TR - fR = I\alpha$
Kaymadan dönme için $a = R\alpha$
$I = \frac{1}{2}MR^2$,
İki denklem arasında $\alpha$ ve $f$ yok edilirse ivme,
$a = (2T - Mg\sin\theta)/\left[M + (I/R^2)\right]$ $\boxed{a = 2(2T - Mg\sin\theta)/3M}$



**PROBLEM (5) (20 points)**

A string is wound around a uniform solid cylinder of radius $R$ (= 10 cm) and mass $M$ (= 16 kg). The cylinder is then unwound under a constant force $F$ (= 60 N). Assume that the cylinder starts from rest and rolls without slipping on the horizontal surface. (The moment of inertia of the cylinder about its central axis is $I = MR^2/2$.)




(a) Calculate the acceleration of the center of mass of the cylinder. (5 points)

$\tau_P = I_P \alpha$

$\tau_P = (2R)(F)$

$I_P = I_{com} + Mh^2$ (Parallel-axis theorem)

$I_P = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$

$\therefore (2R)(F) = \frac{3}{2}MR^2\alpha$ → $\alpha = \frac{4F}{3MR}$

$a_{com} = R\alpha = \frac{4F}{3M} = \frac{4(60\,N)}{3(16\,kg)}$

$\therefore \boxed{a_{com} = 5\ m/s^2}$

OR!

About the center of mass:

$\tau_{com} = RF - Rf_s = I_{com}\alpha$

$(F - f_s)R = \frac{1}{2}MR^2\alpha$

$F - f_s = \frac{1}{2}MR\alpha$

$F - f_s = \frac{1}{2}Ma_{com}$

$+\quad F + f_s = Ma_{com}$ (Newton's 2nd Law)

$\Sigma F = \frac{3}{2}Ma_{com}$

$\therefore \boxed{a_{com} = 5\ m/s^2}$

(b) Calculate the force of static friction $f_s$ necessary to prevent slipping. (5 points)

$F_{net} = Ma_{com} \rightarrow F + f_s = Ma_{com}$

$\therefore f_s = Ma_{com} - F = (16\,kg)(5\,m/s^2) - 60\,N$

$\boxed{f_s = 20\ N}$

(c) Calculate the angular speed of the cylinder after it has rolled through an angle of 25 radians. (5 points)

$\omega^2 = \omega_0^2 + 2\alpha\theta$ ($\omega_0 = 0$)

$\omega^2 = 2\alpha\theta$

$= 2\frac{a_{com}}{R}\theta$

$= 2\frac{5\,m/s^2}{0.1\,m}(25\,rad)$ → $\omega^2 = 2500\ rad^2/s^2$

$\therefore \boxed{\omega = 50\ rad/s}$

OR! The distance traveled by the cylinder is

$d = R\theta = (0.1\,m)(25\,rad) = 2.5\,m$

$v_{com}^2 = v_{com,0}^2 + 2a_{com}d$ ($v_{com,0} = 0$)

$R^2\omega^2 = 2a_{com}d$

$\omega^2 = \frac{2a_{com}d}{R^2} = \frac{2(5\,m/s^2)(2.5\,m)}{0.01\,m^2}$

$\therefore \boxed{\omega = 50\ rad/s}$

(d) Calculate the work done on the cylinder during the turn through 25 radians. (5 points)

The work-KE theorem:

$W = \Delta K = K - K_0$ ($K_0 = 0$)

$K = \frac{1}{2}Mv_{com}^2 + \frac{1}{2}I_{com}\omega^2$

$= \frac{1}{2}M(R\omega)^2 + \frac{1}{2}I_{com}\omega^2$

$= \frac{1}{2}(16\,kg)(0.1 \times 50\,m/s)^2 + \frac{1}{2}\left(\frac{1}{2}(16\,kg)(0.1\,m)^2\right)(50\,rad/s)^2$

$\therefore \boxed{W = 200\ J + 100\ J = 300\ J}$

OR! Take the rolling as a pure rot. about P

$W = \Delta K = \frac{1}{2}I_P\omega^2$

$= \frac{1}{2}\left(\frac{3}{2}MR^2\right)\omega^2$

$\boxed{W = 300\ J}$

OR! If we consider the rolling as a pure rotation about P, then

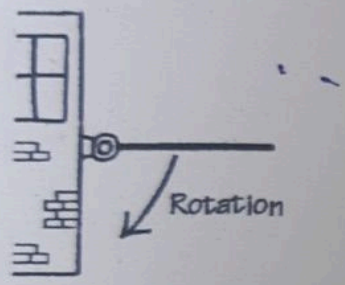
$W = \int_0^{25\,rad} \tau_{com}\,d\theta = (2R)(F)(\theta|_0^{25\,rad})$

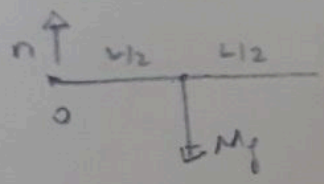
$= (2)(0.1\,m)(60\,N)(25\,rad)$

$\boxed{W = 300\ J}$

**3)**- A uniform rod 1.1 m long with mass 0.7 kg is pivoted at one end, as shown in the figure, and releases from a horizontal position. Answer the following for the moment just after it is released. (a) Find the torque exerted on the rod by the force of gravity. (b) What is the angular acceleration of the rod? (c) What is the acceleration of the center of mass of the rod? (d) What is the net force acting on the rod? (e) What is the magnitude and direction of the force exerted by the pivot? (The moment of inertia of a rod with length $L$ and mass $M$ about an axis going through its center of mass (CM) is $I_{CM} = {}^{1}/_{12}\,ML^2$.)

Kütlesi 0,7 kg, uzunluğu 1,1 m olan homojen bir çubuk şekilde görüldüğü gibi bir ucundan etrafında dönebileceği menteşeye tutturulmuştur. Çubuk yatay konumdan serbest bırakılmıştır. Sorulanları çubuk serbest bırakıldıktan hemen sonraki an için cevaplayınız. (a) Yerçekimi kuvveti tarafından çubuk üzerine etki eden torku bulunuz. (b) Açısal ivmesi nedir? (c) Kütle merkezinin ivmesi nedir? (d) Çubuk üzerine etki eden net kuvvet nedir? (e) Menteşenin çubuğa uyguladığı kuvvetin yönü ve büyüklüğü nedir? ( $M$ kütleli $L$ uzuluğunda çubuğun kütle merkezinden (KM) geçen bir eksene göre eylemsizlik momenti $I_{KM} = {}^{1}/_{12}\,ML^2$ olarak verilir.)




(2)

a) $\tau = Mg\frac{L}{2} = \frac{0{,}7 \cdot 9{,}80 \cdot 1{,}1}{2} = 3.773\ Nm$ (2) (1) ↙ yönde

b) $I = I_{KM} + Md^2 = \frac{1}{12}ML^2 + M\left(\frac{L}{2}\right)^2 = \frac{1}{3}ML^2$ (3)

$\tau = I\alpha$ (2), $\alpha = \frac{\tau}{I} = \frac{Mg\,L/2}{\frac{1}{3}ML^2} = \frac{3g}{2L} = \frac{3\times 9.80}{2\cdot 1{,}1} = 13.36\ \frac{rad}{s^2}$ (1) ↙ yönde

c) $a_{KM} = R\alpha = \frac{L}{2}\cdot\frac{3g}{2L} = \frac{3g}{4} = \frac{3\times 9{,}8}{4} = 7.35\ \frac{m}{s^2}$ (2) (1) ↓ yönde

d) $F_{net} = Ma_{KM} = \frac{3Mg}{4} = 0{,}7 \cdot 7.35 = 5.1415\ N$ (2) (1) ↓ yönde

e) $Mg - n = F_{net}$ (2), $n = Mg - F_{net} = Mg - \frac{3Mg}{4} = \frac{Mg}{4} = \frac{0{,}7\times 9{,}8}{4}$

(1) $n = 1.715\ N$ ↑ yönde